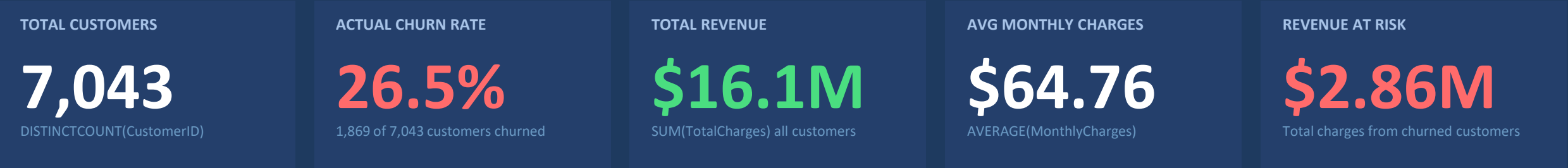


Customer Churn & Retention Analytics

Executive Overview



AI RISK DISTRIBUTION



CHURN vs. RETAINED



MODEL PERFORMANCE (XGBoost Optimised)

ROC-AUC: 0.846

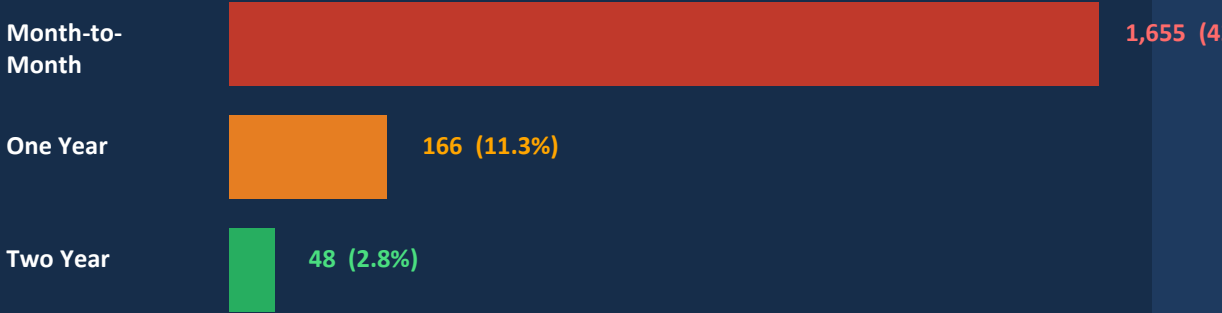
F1 Score: 0.627

Precision: 52.6%

Churn Analysis

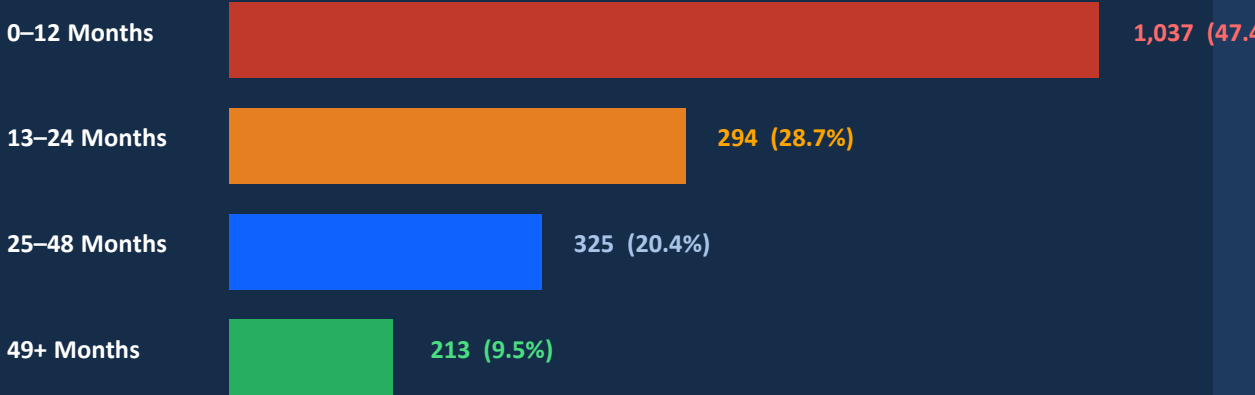
Churn by Contract Type | Tenure Band | Monthly Charges | Payment Method | Internet Service

CHURN BY CONTRACT TYPE

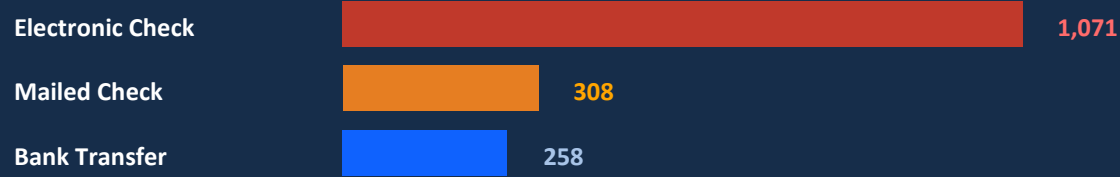


Key Insight: Month-to-Month customers churn at 42.7% vs 11.3% (1-yr) vs 2.8% (2-yr). Contract type is a primary churn driver.

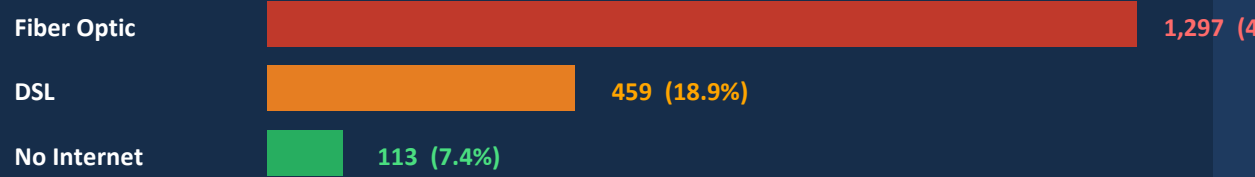
CHURN BY TENURE BAND



CHURN BY PAYMENT METHOD

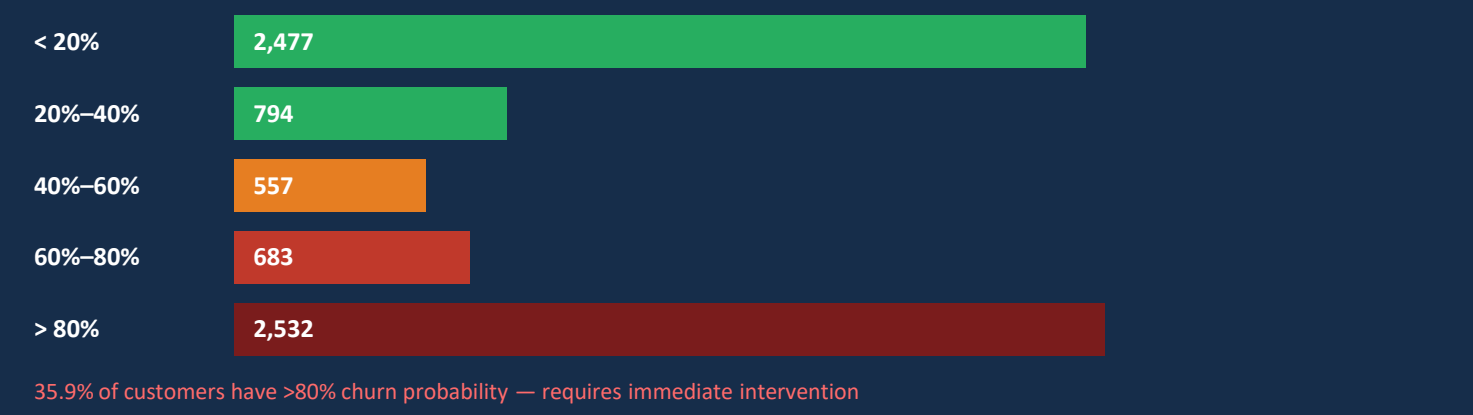


CHURN BY INTERNET SERVICE



AVG MONTHLY CHARGES: Churned \$74.44 vs Retained \$61.27 (+21.5% higher for churned customers)

CHURN PROBABILITY DISTRIBUTION (7,043 customers)



RISK SEGMENT KPIs

HIGH RISK
3,215 customers
45.7% | Prob > 60%
Actual Churn: 1,533

MEDIUM RISK
912 customers
12.9% | Prob 40–60%
Actual Churn: 241

LOW RISK
2,916 customers
41.4% | Prob < 40%
Retained: 2,821

PREDICTED CHURN
3,294 customers
Recall: 79.1%
Precision: 52.6%

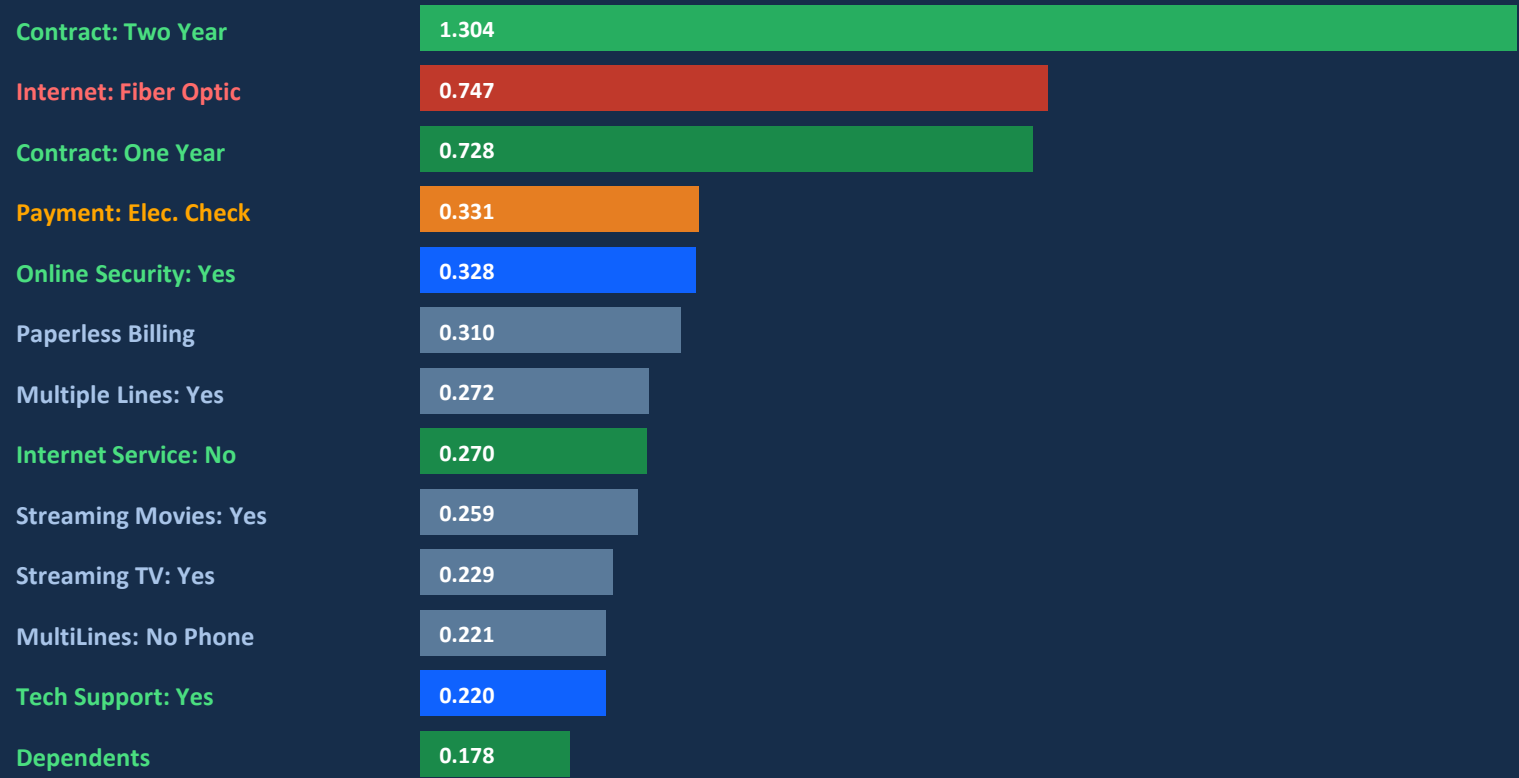
TOP 10 HIGH-RISK CUSTOMERS (by Churn Probability)

Customer ID	Churn Prob	Risk	Contract	Monthly \$	Top Recommendation
7359-SSBJK	99.5%	High Risk	Two year	\$70.20	Senior discount + loyalty points programme
4439-YRNVD	99.5%	High Risk	Month-to-month	\$36.25	Discounted 1-yr contract + security bundle
2812-REYAT	99.5%	High Risk	One year	\$20.05	Proactive satisfaction call
9369-YEEHK	99.5%	High Risk	Month-to-month	\$80.60	Contract upgrade +

Explainable AI — SHAP Feature Importance

SHapley Additive exPlanations (SHAP) | XGBoost Model | 500 Customers Sample | Mean |SHAP| Values

TOP 15 CHURN DRIVERS (Mean |SHAP| Value — higher = more impact on churn prediction)



KEY SHAP INSIGHTS

- #1 RETENTION DRIVER

Two-Year Contract is the strongest churn-prevention signal (SHAP=1.304). Customers locked into long-term contracts are 15x less likely to churn.
- #2 CHURN ACCELERATOR

Fiber Optic internet strongly increases churn risk (SHAP=0.747). Fiber users may perceive value-price mismatch despite premium pricing.
- #3 PAYMENT SIGNAL

Electronic check payers have 57.3% of all churns. This payment method correlates with month-to-month contracts and lower engagement.
- #4 SECURITY SERVICES

Having Online Security & Tech Support significantly reduces churn risk. Bundling these services is a proven retention strategy.